

Day 6 Plan

Maths: LCM & HCF, Simplification, Surds · Reasoning: Blood Relations · Science:
Physics 6: Light

Today's schedule

#	Block	Time	What you do	Where
0	Day 5 recall	15 min	10 questions from Day 5, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	Read the formulas & methods (10 min) → 2-3 book questions per pattern, timed → log mistakes	Section A
-	Break	10 min	Away from the screen	-
2	Reasoning	50 min	Learn the method (10 min) → 15 timed questions from your book → log	Section B
-	Break	10 min	Away from the screen	-
3	Science	60 min	Lessons with quick checks → 20-question practice set → answers	Section C
4	Current affairs	20 min	20 MCQs from the monthly PDF	Your PDF
5	Recall	10 min	Redo today's wrong questions; write 3 rules for tomorrow	Mistake log

Speed targets: Maths 60-70 sec per question · Reasoning 30-40 sec · Science 15-20 sec.

Block 0 — Day 5 recall (answers on the last page)

#	Question	Your answer
1	Sound cannot travel through ...?	
2	Fastest medium for sound: air, water or steel?	
3	Sound waves in air are transverse or longitudinal?	
4	Pitch depends on ...?	
5	Human audible range?	
6	Bats and SONAR use which waves?	
7	Minimum distance for an echo ($v = 340 \text{ m/s}$)?	
8	'Supersonic' describes what?	
9	Approaching train whistle sounds higher — which effect?	
10	Maths: unit digit of 3^{48} ?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — LCM & HCF, Simplification, Surds

Core idea

HCF (highest common factor) = the biggest number that divides all the given numbers — it's about **splitting into equal groups**. **LCM** (lowest common multiple) = the smallest number all of them divide into — it's about things **happening together again** (bells, lights, runners).
With prime factors: HCF takes the **lowest** powers, LCM the **highest**.

Formulas

Concept	Formula / rule
HCF / LCM by prime factors	$12 = 2^2 \times 3$, $18 = 2 \times 3^2 \rightarrow \text{HCF} = 2 \times 3 = 6$ · $\text{LCM} = 2^2 \times 3^2 = 36$
Two numbers only	HCF × LCM = product of the two numbers
Fractions	$\text{HCF} = \text{HCF}(\text{numerators})/\text{LCM}(\text{denominators})$ · $\text{LCM} = \text{LCM}(\text{numerators})/\text{HCF}(\text{denominators})$
Largest number leaving remainders r_1, r_2, r_3	HCF of $(a - r_1), (b - r_2), (c - r_3)$
Largest number leaving the SAME remainder	HCF of the differences $(b - a), (c - b), (c - a)$
Smallest number leaving remainder r	$\text{LCM} + r$
Greatest n-digit number divisible by a, b, c	Largest n-digit number — (its remainder when divided by the LCM)
BODMAS	Brackets $\rightarrow$ of $\rightarrow \div \rightarrow \times \rightarrow + \rightarrow -$ ('of' is done before division)
Surds	$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ · $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$ · $1/(\sqrt{a} + \sqrt{b}) = (\sqrt{a} - \sqrt{b})/(a - b)$
Square root of $a \pm 2\sqrt{b}$	$= \sqrt{x} \pm \sqrt{y}$ where $x + y = a$ and $xy = b$

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Basic HCF/LCM	Prime factorise	12 and 18 $\rightarrow$ HCF 6 , LCM 36
2. Find the other number	$\text{HCF} \times \text{LCM} \div \text{given number}$	HCF 12, LCM 360, one number 72 $\rightarrow 12 \times 360 \div 72 =$ 60
3. Largest divisor with remainder	HCF of (numbers – remainder)	Divides 70, 100, 130 leaving 10 $\rightarrow \text{HCF}(60, 90, 120) =$ 30
4. Smallest number with remainder	$\text{LCM} + r$	Divided by 6, 8, 12 leaves 5 $\rightarrow 24 + 5 =$ 29
5. Bells / lights together	LCM	Bells every 6, 8, 12 min $\rightarrow$ together again after 24 min
6. Same remainder	HCF of differences	43, 91, 183 $\rightarrow \text{HCF}(48, 92, 140) =$ 4 (each leaves 3 ✓)
7. Greatest 4-digit multiple	Use the LCM	Divisible by 12, 15, 20 $\rightarrow \text{LCM } 60 \rightarrow 9999 \div 60$ leaves 39 $\rightarrow$ 9960
8. Fractions	Formulas above	$2/3$ and $4/9 \rightarrow$ HCF $2/9$, LCM $4/3$
9. BODMAS	'of' before $\div$	$18 \div 3$ of $2 + 4 \times 2 \rightarrow 18 \div 6 + 8 =$ 11
10. Rationalise	Multiply by the conjugate	$1/(\sqrt{3} + \sqrt{2}) =$ $\sqrt{3} - \sqrt{2}$
11. Nested root	Find $x + y = a, xy = b$	$\sqrt{5 + 2\sqrt{6}} \rightarrow 3 + 2 = 5, 3 \times 2 = 6 \rightarrow$ $\sqrt{3} + \sqrt{2}$

Traps

- 'HCF $\times$ LCM = product' works for **two** numbers only.
- 'Of' is calculated **before** division: $18 \div 3$ of $2 = 18 \div 6$, not 6×2 .
- HCF is always $\leq$ the smallest number; LCM is always $\geq$ the largest. Use this to eliminate options instantly.

B · REASONING — Blood Relations

Always draw a family tree. Use + for male and – for female (or $\square$ and $\circ$), a horizontal double line for a married couple, a vertical line for parent $\rightarrow$ child, and a single horizontal line for siblings. Same row = same generation.

Relation	Means
Father's or mother's brother	Uncle (paternal / maternal)
Father's or mother's sister	Aunt
Uncle's or aunt's child	Cousin
Brother's / sister's son or daughter	Nephew / niece
Son's wife / daughter's husband	Daughter-in-law / son-in-law
Husband's or wife's brother; sister's husband	Brother-in-law
Husband's or wife's sister; brother's wife	Sister-in-law
Father's or mother's father	Grandfather (paternal / maternal)
'Only son of my grandfather'	My father (when it's the paternal grandfather)
'Only daughter of my mother'	Me, if the speaker is female

Patterns, with solved examples

Pattern	Example	Solution
Photograph	Pointing to a photo, Ravi says: 'He is the son of the only son of my grandfather.'	Only son of my grandfather = my father; his son (not me) $\rightarrow$ Ravi's brother
Chain	A is B's sister. C is B's mother. D is C's father. A is D's ...?	A is C's daughter $\rightarrow$ D's granddaughter
Coded	A + B: A is B's mother · A – B: A is B's brother · A $\times$ B: A is B's father. P $\times$ Q – R means?	P is Q's father, Q is R's brother $\rightarrow$ P is R's father
Introduction	A woman says of a man: 'His mother is the only daughter of my mother.'	Only daughter of my mother = herself $\rightarrow$ she is his mother
Cousin	X's mother's brother's son is X's ...?	Cousin
Can't tell	A is B's brother. C is A's sibling. How is C related to B?	C's gender is unknown $\rightarrow$ brother or sister (cannot be determined)

Speed rule

Never assume gender from a name. If the question doesn't state it and no option depends on it, look for 'cannot be determined'.

C · SCIENCE — Physics 6: Light

9 lessons — light is one of the highest-scoring physics topics. Draw each diagram once in your rough book; it helps you remember mirror and lens uses.

Lesson 1 · Nature of light (about 4 min)

Light is an **electromagnetic wave**: it needs no medium and travels at **3×10^8 m/s in vacuum** — the fastest speed possible. It slows down in materials: $\approx 2.25 \times 10^8$ m/s in water, $\approx 2 \times 10^8$ m/s in glass. Sunlight takes about **8 minutes 20 seconds** to reach Earth.

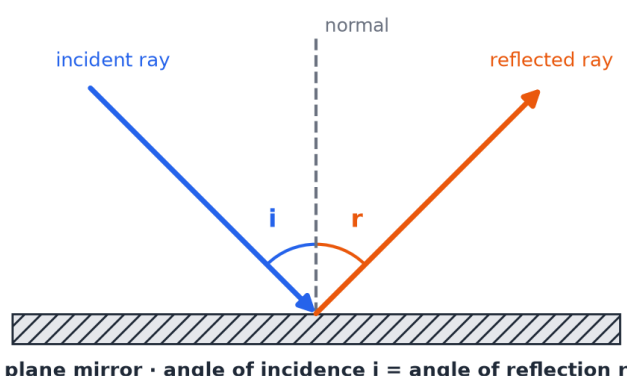
Light travels in **straight lines** (rectilinear propagation). That's why shadows and eclipses form, and why a **pinhole camera** gives a **real, inverted** image.

Term	Meaning / example
Luminous / non-luminous	Gives off its own light (Sun, bulb) / doesn't (Moon, book)
Transparent	Lets almost all light through — clear glass, water
Translucent	Lets some light through — butter paper, frosted glass
Opaque	Lets no light through — wood, metal

Quick check 1 — answer out loud before moving on (answers at the end)

a) What is the speed of light in vacuum? b) What kind of image does a pinhole camera form?

Lesson 2 · Reflection and plane mirrors (about 5 min)



Laws of reflection: (1) angle of incidence = angle of reflection; (2) the incident ray, the reflected ray and the normal lie in the same plane.

Plane mirror image	Detail
Virtual and erect	Can't be caught on a screen; upright
Same size, same distance behind the mirror	Stand 2 m in front → the image is 2 m behind (4 m from you)
Laterally inverted	Left ↔ right — why 'AMBULANCE' is written in reverse on ambulances
Mirror length to see yourself fully	At least half your height
Two mirrors at angle θ	Number of images = $360/\theta - 1$ ($90^\circ \rightarrow 3$ · $60^\circ \rightarrow 5$ · parallel mirrors → infinite)

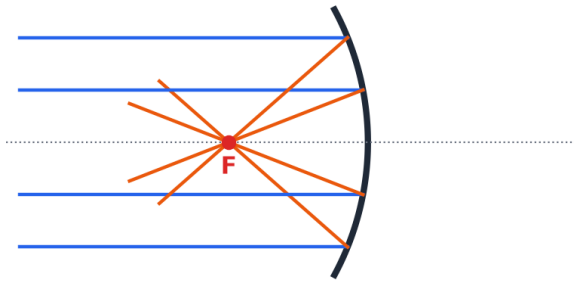
Quick check 2 — answer out loud before moving on (answers at the end)

How many images are formed by two plane mirrors at 60° to each other?

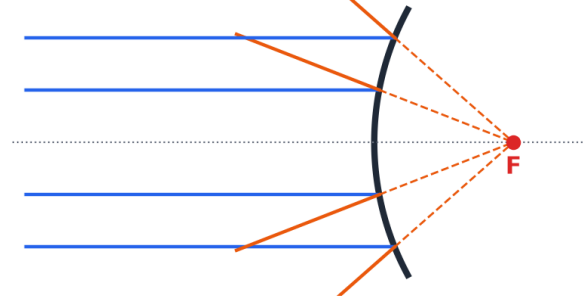
Lesson 3 · Concave and convex mirrors (about 8 min)

A spherical mirror is part of a hollow sphere. **Concave** = reflecting surface curved inward (like the inside of a spoon). **Convex** = curved outward (the back of a spoon). The **focus F** is where parallel rays meet (or seem to come from). **Focal length $f = R/2$** (R = radius of curvature).

CONCAVE (converging)
parallel rays meet at the focus F



CONVEX (diverging)
rays spread out, seem to come from F behind



Object position (concave mirror)	Image
At infinity (very far)	At F — real, inverted, point-sized
Beyond C	Between F and C — real, inverted, smaller
At C	At C — real, inverted, same size
Between C and F	Beyond C — real, inverted, larger
At F	At infinity
Between F and the mirror	Behind the mirror — virtual, erect, enlarged

A **convex mirror** always gives a **virtual, erect, smaller** image, but shows a **wide area**.

Concave mirror uses	Convex mirror uses
Shaving and make-up mirrors (enlarged, upright face) · dentist's mirror · torch, headlight and searchlight reflectors (bulb at F → parallel beam) · solar cookers and furnaces	Rear-view (side) mirrors of vehicles (wide view, upright) · security mirrors in shops · mirrors at sharp road bends · street-light reflectors

Worked example — focal length

A concave mirror has radius of curvature 30 cm → $f = 30 \div 2 = 15$ cm.

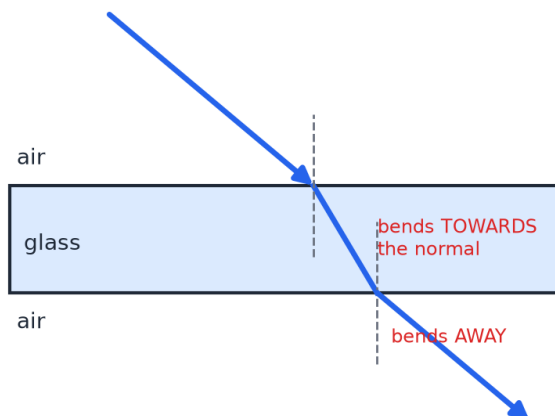
Quick check 3 — answer out loud before moving on (answers at the end)

a) Which mirror is used as a rear-view mirror, and why? b) Where is the bulb placed in a headlight reflector?

Lesson 4 • Refraction (about 6 min)

Refraction is the bending of light when it passes from one medium into another, because its **speed changes**. Going into a **denser** medium (air → glass), light slows down and bends **towards the normal**; going into a rarer medium, it bends **away**.

air → glass: light slows down and bends towards the normal



Refractive index n = speed of light in vacuum ÷ speed in the medium. Water ≈ 1.33 , glass ≈ 1.5 , **diamond ≈ 2.42** (very high).

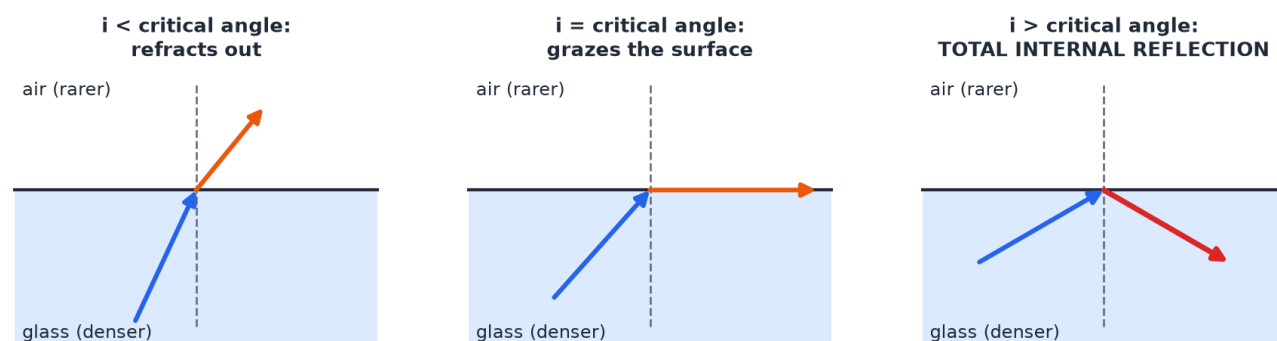
Everyday effect of refraction	
A swimming pool or river looks shallower than it is	
A pencil in a glass of water looks bent	
Stars twinkle (atmospheric refraction; planets don't twinkle much)	
We see the Sun about 2 minutes before actual sunrise and after actual sunset	

Quick check 4 — answer out loud before moving on (answers at the end)

a) Why does a pool look shallower than it really is? b) Why do stars twinkle?

Lesson 5 • Total internal reflection (TIR) (about 5 min)

When light goes from a **denser to a rarer** medium and hits the surface at an angle **greater than the critical angle**, none of it escapes — it is **totally reflected** back inside.



Example of TIR	How
Sparkling of diamond	Very small critical angle ($\approx 24^\circ$), so light bounces inside many times before coming out
Optical fibres	Light zig-zags along the fibre by TIR — internet cables, endoscopes

Example of TIR	How
Mirage (water seen on a hot road or desert)	Hot air near the ground is rarer; light from the sky is totally reflected upward

Quick check 5 — answer out loud before moving on (answers at the end)

Name two everyday uses or examples of total internal reflection.

Lesson 6 · Lenses and power (about 5 min)

Convex lens (thick in the middle) = **converging**. **Concave lens** (thin in the middle) = **diverging**.

Power of a lens $P = 1/f$ (f in **metres**), unit **diopetre (D)**. Convex lens → **positive** power; concave → **negative**. Lenses in contact: $P = P_1 + P_2$.

Worked example — power

Convex lens, $f = 50 \text{ cm} = 0.5 \text{ m} \rightarrow P = 1/0.5 = +2 \text{ D}$.

Concave lens, $f = 25 \text{ cm} \rightarrow P = -1/0.25 = -4 \text{ D}$.

+2 D and -4 D in contact → **-2 D** (acts as a diverging lens).

Convex lens uses	Concave lens uses
Magnifying glass · camera · eye lens · microscope · telescope · spectacles for hypermetropia	Spectacles for myopia · wide-angle door viewers (peepholes)

Quick check 6 — answer out loud before moving on (answers at the end)

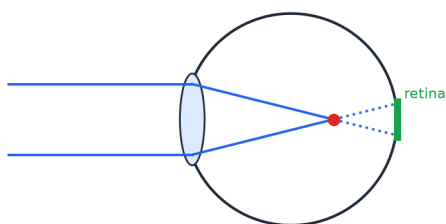
What is the power of a convex lens of focal length 20 cm?

Lesson 7 · The human eye and its defects (about 7 min)

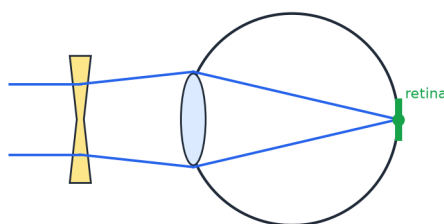
Part	Job
Cornea	Transparent front window — does most of the bending of light
Iris	Coloured part — controls the size of the pupil
Pupil	Opening that lets light in (small in bright light, large in dim light)
Lens	Convex lens; ciliary muscles change its focal length so we can focus near and far (accommodation)
Retina	Screen with light-sensitive cells: rods (dim light) and cones (colour)
Optic nerve	Carries signals to the brain · the blind spot is where it leaves the eye (no rods or cones)

The image on the retina is **real and inverted**; the brain turns it upright. Normal eye: near point **25 cm**, far point **infinity**. **Persistence of vision** ($\approx 1/16 \text{ s}$) is why a series of still frames looks like a moving film.

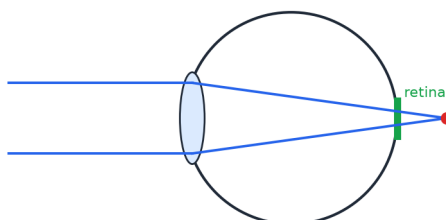
MYOPIA: image forms IN FRONT of the retina



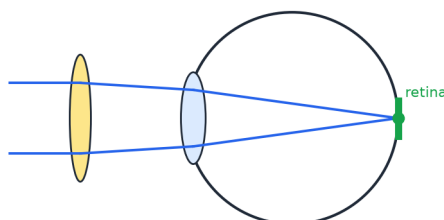
Corrected with a CONCAVE lens



HYPERMETROPIA: image would form BEHIND the retina



Corrected with a CONVEX lens



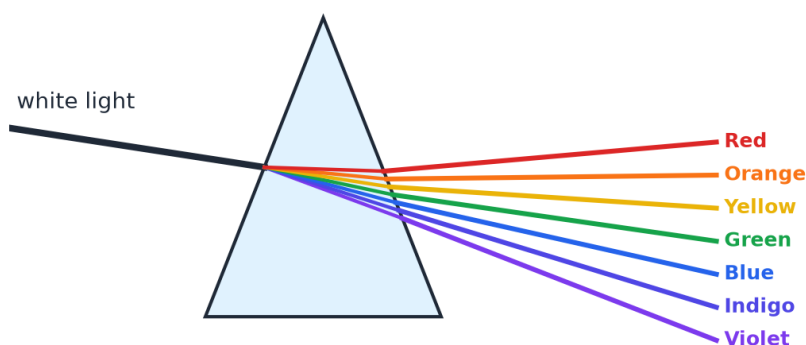
Defect	Problem	Correction
Myopia (short-sightedness)	Can't see far objects clearly; image forms in front of the retina	Concave lens
Hypermetropia (long-sightedness)	Can't see near objects clearly; image would form behind the retina	Convex lens
Presbyopia	Old age: lens loses flexibility; both near and far may blur	Bifocal lenses
Astigmatism	Unequal curvature of the cornea	Cylindrical lens
Cataract	Eye lens becomes cloudy	Surgery (lens replaced)

Quick check 7 — answer out loud before moving on (answers at the end)

a) Which lens corrects myopia? b) What kind of image forms on the retina?

Lesson 8 · Dispersion, spectrum and rainbow (about 5 min)

Newton showed that white light passing through a **prism** splits into seven colours — **dispersion**. Each colour has a different wavelength and bends by a different amount.



Red bends LEAST (longest wavelength) · violet bends MOST · VIBGYOR

VIBGYOR: violet, indigo, blue, green, yellow, orange, red. **Red** has the longest wavelength and bends the **least**; **violet** has the shortest and bends the **most**.

Rainbow = dispersion + refraction + internal reflection inside raindrops. You see it only with the **Sun behind you**. **Primary colours of light**: red, green, blue (RGB) → together they make

white. Secondary: yellow, cyan, magenta.

Quick check 8 — answer out loud before moving on (answers at the end)

a) Which colour is deviated least by a prism? b) Where must the Sun be for you to see a rainbow?

Lesson 9 · Scattering and colours of objects (about 5 min)

Tiny particles in the air scatter **short wavelengths (blue)** much more than long ones (red).

Observation	Explanation
The sky is blue	Blue light is scattered most by air molecules and reaches us from every direction
Sunrise and sunset look red	Sunlight crosses much more air, so most blue is scattered away; red gets through
Danger signals are red	Red is scattered least, so it can be seen from far away, even in fog
The sky looks black to astronauts in space	No atmosphere, so no scattering
Clouds are white	Large water droplets scatter all colours roughly equally
A beam is visible in a dusty room (Tyndall effect)	Dust particles scatter the light sideways

Colour of an object = the colour it **reflects**. A red rose reflects red and absorbs the rest — so under **green light** it looks **black**.

Quick check 9 — answer out loud before moving on (answers at the end)

a) Why is the sky blue? b) Why are danger signals red?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Light basics	EM wave · 3×10^8 m/s in vacuum (slower in water, glass) · Sun to Earth ≈ 8 min 20 s · straight lines → shadows, pinhole camera (inverted)
Plane mirror	$i = r$ · virtual, erect, same size, laterally inverted · half-height mirror · images = $360/\theta - 1$
Concave mirror	Converging · shaving/make-up, dentist, headlights (bulb at F), solar cooker · virtual enlarged image only when object is inside F
Convex mirror	Diverging · always virtual, erect, smaller, wide view · rear-view mirrors, shop security
Focal length	$f = R/2$
Refraction	Into denser → towards the normal · $n = c/v$ · pool looks shallow, pencil bent, stars twinkle, early sunrise
TIR	Denser → rarer, angle > critical · diamond sparkle, optical fibre, mirage
Lenses	Convex converging (+), concave diverging (−) · $P = 1/f$ (m), dioptre · $P = P_1 + P_2$
Eye	Cornea, iris, pupil, lens (accommodation), retina (rods dim, cones colour) · image real, inverted · near point 25 cm
Defects	Myopia → concave · hypermetropia → convex · presbyopia → bifocal · astigmatism → cylindrical · cataract → surgery
Dispersion	VIBGYOR · red least deviated, violet most · rainbow: Sun behind you · RGB primary
Scattering	Blue sky · red sunset · red danger signals · black sky in space · white clouds · red rose in green light looks black

Science practice set — 20 questions · target 6 minutes

#	Question
1	The speed of light in vacuum is: (a) 3×10^6 m/s (b) 340 m/s (c) 3×10^8 m/s (d) 3×10^5 m/s
2	The image formed by a plane mirror is: (a) virtual, erect and laterally inverted (b) real and inverted (c) real and erect (d) virtual and inverted
3	The number of images formed by two plane mirrors at 90° to each other is: (a) 4 (b) 3 (c) 2 (d) infinite
4	The rear-view mirror of a vehicle is a: (a) concave mirror (b) plane mirror (c) cylindrical mirror (d) convex mirror
5	A shaving mirror is a: (a) concave mirror (b) convex mirror (c) plane mirror (d) convex lens
6	In a car headlight, the bulb is placed at the reflector's: (a) centre of curvature (b) focus (c) pole (d) anywhere in front
7	The focal length of a mirror with radius of curvature 30 cm is: (a) 30 cm (b) 60 cm (c) 15 cm (d) 7.5 cm
8	A swimming pool appears shallower than it is because of: (a) refraction (b) reflection (c) dispersion (d) scattering
9	The sparkling of a diamond is due to: (a) dispersion only (b) total internal reflection (c) scattering (d) diffraction
10	Optical fibres work on the principle of: (a) refraction only (b) dispersion (c) total internal reflection (d) scattering
11	A mirage in a desert is caused by: (a) scattering (b) total internal reflection (c) dispersion (d) diffraction
12	The power of a convex lens of focal length 50 cm is: (a) +2 D (b) -2 D (c) +0.5 D (d) +50 D
13	Myopia (short-sightedness) is corrected with a: (a) convex lens (b) cylindrical lens (c) bifocal lens (d) concave lens
14	Hypermetropia (long-sightedness) is corrected with a: (a) convex lens (b) concave lens (c) cylindrical lens (d) plane glass
15	The image formed on the retina is: (a) virtual and erect (b) real and erect (c) real and inverted (d) virtual and inverted
16	Which colour is deviated least by a prism? (a) red (b) violet (c) green (d) blue
17	The sky appears blue because of: (a) reflection of light (b) dispersion of light (c) scattering of light (d) refraction of light
18	Danger signals are red because red light: (a) is scattered least (b) is scattered most (c) travels fastest (d) is easily absorbed
19	Stars twinkle because of: (a) reflection (b) scattering (c) atmospheric refraction (d) total internal reflection
20	A rainbow can be seen when: (a) the Sun is behind the observer (b) the Sun is in front of the observer (c) the Sun is overhead (d) it is night

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	c	Lesson 1
2	a	Lesson 2
3	b	$360/90 - 1$ (Lesson 2)
4	d	Lesson 3
5	a	Lesson 3
6	b	Lesson 3
7	c	$f = R/2$ (Lesson 3)
8	a	Lesson 4
9	b	Lesson 5
10	c	Lesson 5
11	b	Lesson 5
12	a	$1/0.5 \text{ m}$ (Lesson 6)
13	d	Lesson 7
14	a	Lesson 7
15	c	Lesson 7
16	a	Lesson 8
17	c	Lesson 9
18	a	Lesson 9
19	c	Lesson 4
20	a	Lesson 8

Quick checks

QC	Answers
1	a) $3 \times 10^8 \text{ m/s}$ b) real and inverted
2	$360/60 - 1 = 5$
3	a) Convex — gives an erect image and a wide field of view b) At the focus, so the reflected beam is parallel
4	a) Refraction: light from the bottom bends away from the normal as it leaves the water b) Atmospheric refraction
5	Any two: sparkling of diamond, optical fibres, mirage
6	$f = 0.2 \text{ m} \rightarrow P = +5 \text{ D}$
7	a) Concave lens b) Real and inverted
8	a) Red b) Behind you
9	a) Air scatters blue light the most b) Red is scattered least, so it is visible from far away

Day 5 recall (Block 0)

#	Answer	#	Answer
1	Vacuum	6	Ultrasonic
2	Steel	7	17 m
3	Longitudinal	8	Speed greater than the speed of sound
4	Frequency	9	Doppler effect
5	20 Hz - 20 kHz	10	$48 \div 4$ leaves 0 $\rightarrow$ 4th term of 3, 9, 7, 1 $\rightarrow$ 1

Send me tonight

Day 5 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.